MyMarket

An MCP server that reads Malaysian news and tells a food buyer whether staple prices are about to move, and what to do about it

Version	0.4, food staples only
Document built	2026-10-11
Domain	food_staples: rice, egg, cooking oil, chicken
Price truth	PriceCatcher, KPDN via data.gov.my, daily, CC BY 4.0
News	153,981 articles, 2020-10-31 to 2026-10-09
Interface	Model Context Protocol, 9 tools, stdio and HTTP
Code	49 Python files, 8,140 lines, 145 tests passing
Headline result	macro F1 0.333 against 0.315 for the best baseline
Alert quality	20 alerts at confidence 0.70 or above, precision 0.70

This document is generated from the repository. Every number in it is read at build time from `models/metrics.json`, the processed data and the config. Re-run `scripts/09_build_project_doc.py` after retraining and it stays true.

1. What this is and who it is for

A kopitiam owner buys chicken every week. A caterer buys rice by the sack. If prices are about to rise they want to buy early; if prices are about to ease they want to wait. Today they find out when the invoice arrives.

MyMarket reads Malaysian news as it is published, keeps only the stories genuinely about four food staples, predicts whether the shelf price will move over the next 28 days, checks that call against the government's own daily price survey, ranks it by urgency, and says what to do. It is exposed as an MCP server, so the whole thing is usable from a chat client.

It is an awareness tool, not a forecast, and not financial advice. That framing is in the product, not just the pitch: every alert carries the disclaimer, the model abstains when unsure, and the accuracy is published alongside the baselines it has to beat.

2. Scope

The original specification covered two domains. Consumer electronics was cut in v0.3 because no free Malaysian dataset carries product level electronics prices; the fallback was monthly CPI proxy classes, which give roughly 60 independent labels and cannot support an honest claim. Food staples have a daily, item level, official price survey. Dropping electronics removed the DOSM CPI pipeline, the monthly labelling path, and PRD flags F2, F14 and F15.

3. Data

Source	What it gives	Volume
PriceCatcher records storage.data.gov.my	Daily price per item per premise, reduced offline to a daily table	80,437 daily rows 2022-01-01 to 2026-09-24 122 items
PriceCatcher lookups	item_code and premise_code to names and categories	797 items
Kaggle news dump azraimohamad/news-article-weekly-updated	Training news. 100 percent thestar.com.my, which is a source shift against live feeds	153,981 articles 2020-10-31 to 2026-10-09
Live RSS	Malay Mail, Bernama, Free Malaysia Today	about 108 unique per scan

Raw monthly PriceCatcher files hold over a million rows each. They are downloaded one month at a time, reduced to a daily aggregate and deleted, so peak disk stays near a single month and the server never opens a raw file.

4. The pipeline, end to end

Stage	Script	What it does
1 Prices	01_download_prices.py	Probes every month from 2022-01, downloads, aggregates to daily median, p25, p75 and premise count, deletes the raw file, resumes if interrupted
2 Mapping	04_map_items.py	Matches PriceCatcher items to products, scores each candidate by history length and survey coverage, writes a review sheet and a provisional mapping
3 Index	03_build_price_index.py	Builds the normalised daily price index per product and a plot for each
4 News	02_prepare_news.py	Cleans the Kaggle dump, writes the audit sample, tunes the filter threshold on the hand labelled result
5 Examples	05_build_examples.py	Joins news to prices, assigns labels, splits by time with a purge gap, prints proof that the gap holds

Stage	Script	What it does
6 Train	06_train_local.py	Trains a TF-IDF baseline and two candidate models, selects on validation, calibrates if calibration helps, scores the test set once
7 Evaluate	07_evaluate.py	Per product metrics, bootstrap intervals, three baselines, rolling origin backtest
8 Demo	08_demo_cases.py	Picks held out cases where the real outcome is known, including ones the model got wrong
Daily	daily_refresh.py	Refreshes prices, logs live predictions, scores matured ones for forward validation

Running order matters: 06 rewrites metrics.json from scratch, so 07 must run after it or the per product and rolling origin sections are lost.

5. Feature: the relevance filter

The hardest problem in this project is not prediction, it is deciding which articles are even about a staple price. The Star publishes far more recipes, buffets and restaurant reviews mentioning chicken than it publishes chicken price stories. Three stages run in order, cheapest first.

Stage	Rule	Measured precision
1 Alias match	Word boundary match on English and Malay aliases across title, summary and the first 1500 characters of body. Exclude phrases are masked out of the text rather than rejecting the article, so "chicken rice stalls hit as chicken supply tightens" still matches	0.175
2 Price cue	The headline or summary must contain one of 28 cue words: price, harga, subsidy, tariff, shortage, inflation and so on. The body is deliberately not searched, because a review mentions cost in paragraph nine while a price story says so in the headline	0.800
3 Embedding	Contrastive score with multilingual-e5-small: best similarity to the product description sentences minus best similarity to 11 distractor sentences. Threshold 0.015	0.875 combined

Why contrastive and not a plain cosine. The embedding model scores anything vaguely on topic in the 0.82 to 0.88 band. Measured directly, "Best fried chicken restaurants in Kuala Lumpur" scored 0.8828 while "Chicken prices rise as feed costs climb" scored 0.8819, so a plain similarity ranks the noise above the signal. Subtracting the distractors separates them.

The biggest false positive, found by audit. Crude palm oil futures, plantation output and biodiesel stories dominated the survivors. They are upstream commodity news, not a retail shelf price, so distractor sentences for them were added.

5.1 Honest note on the filter numbers

Two audits of 120 rows each were hand labelled. Round one was labelled in house and the distractor sentences were tuned on it, which produced 0.700 precision. A second set, labelled independently by a teammate using a different pipeline, scored the same configuration at **0.500**. The 0.700 was an overfit and did not survive. Real precision is around 0.55 at the shipped threshold, rising to 0.80 with the price cue stage. **The 0.80 target in the original PRD is not met; the agreed MVP bar of 0.70 is met only by the cue stage, not by the embedding stage alone.**

6. How the price index is built

A unitless index per product, so a 10 kg rice sack and a 1 kg oil packet can be averaged without the sack dominating.

```
1. daily median price per item across premises, dropping price <= 0
2. drop any item-day outside [0.333, 3.0] x that item's rolling 30 day median
3. normalise each item by its own median over its first 90 days
4. carry each item forward up to 10 days, then average across items, then a 7 day rolling
mean
```

Step 4 is not in the original specification. It was added after the survey cadence was found to degrade badly: by 2026 cooking oil items are surveyed about weekly while eggs remain near daily. Without the carry forward the cooking oil and rice indexes simply stopped in 2025. A survey price is a step function, so the last observed price is the honest estimate until the next survey.

Result: 6,902 index rows covering 1,721 to 1,728 days per product.

What the index shows. Rice rises about 45 percent in September 2023, which is the real import crisis following India's export ban. Egg is a flat line until mid 2024 because eggs were administratively priced. Both are genuine and both are visible in the

plots.

7. How labels are calculated

```
base = I_p(d - 1)
fut = I_p(d + H) H = 28 days
ret = fut / base - 1
theta_p = max(0.005, k * std(ret over TRAIN only)) k = 0.5

ret > theta_p -> price UP shown as NEGATIVELY AFFECTED
ret < -theta_p -> price DOWN shown as POSITIVELY AFFECTED
otherwise -> STABLE shown as NEUTRAL
```

theta is computed on the training period only. Computing it on all data would leak the test period's volatility into the labels. The floor of 0.005 exists because a price controlled product has an almost flat training period, which drives k times std towards zero and would make every later rounding wiggle a move. Any product sitting on the floor is reported, because there the floor and not the data is deciding the labels.

An example with no price window is dropped, never guessed.

8. The model

Element	Choice	Why
Text features	multilingual-e5-small embeddings, 384 dimensions	Handles English and Malay in one model, which matters for Malaysian news
Other features	product one hot, 26 binary event word flags	Policy words move staple prices in steps, so they are features
Products	merged into one group	Per product features lost on validation, 0.440 against 0.242. A few hundred examples cannot support per product parameters
Candidates	TF-IDF baseline, logistic regression, LightGBM	Selected on validation only; the test set is not used to choose
Calibration	rejected, it dropped validation macro F1 from 0.466 to 0.403 on 35 rows, so probabilities	Kept only when it does not hurt validation. On a small imbalanced split it can collapse the model onto the majority class
Abstention	below confidence 0.55	An abstention is not a claim, so it is not logged as a prediction

Examples	Train	Validation	Test	Purged
305	210	35	40	20

Class balance: NEUTRAL 156, NEGATIVELY AFFECTED 70, POSITIVELY AFFECTED 79.

The split is by time with a purge gap, never random. One news event produces many articles sharing a label, so a random split would leak. A 28 day gap is purged at each boundary because an article published just before the boundary is still looking forward across it. The build prints proof of the gap every run.

9. Scoring: what is measured and why

9.1 Why macro F1 and not accuracy

Macro F1 is the unweighted average of the F1 score for each of the three classes, so every class counts equally no matter how rare. Accuracy would reward doing nothing: a model that always says NEUTRAL scores **65 percent accuracy** on this test set because 26 of 40 cases genuinely are NEUTRAL, while never warning anyone about anything. Macro F1 exposes it, scoring that model 0.263.

```
MyMarket accuracy 50.0% macro F1 0.333
Always say NEUTRAL accuracy 65.0% macro F1 0.263
```

Macro F1 is a score from 0 to 1, not a percentage of correct answers. Quoting 0.333 as 33 percent invites the reader to hear "right a third of the time", which is wrong in both directions: accuracy is 50 percent and high confidence alerts are right 70 percent of the time.

9.2 Results against the baselines

Model or baseline	Macro F1	What it is
MyMarket	**0.333**	the trained model
Momentum	0.315	sign of the price move over the 28 days before publication
Keyword rules	0.310	hand written rules on event words
Majority class	0.263	always predict the most common label

95 percent bootstrap interval **[0.218, 0.449]** on 40 test rows. The model beats all three baselines, but it is 0.018 ahead of momentum, which is inside the noise. **Call it a tie trending the right way, not a win.**

Alert precision: 0.70 on 20 alerts at confidence 0.7 or above. This is the number the product actually acts on and the most useful single figure for a non technical reader.

9.3 Per product

Product	Test rows	Macro F1	Directional accuracy
chicken	12	0.429	0.50
cooking oil	8	0.256	0.00
egg	8	0.083	0.14
rice	12	0.246	no moved rows

Directional accuracy counts only the rows that actually moved, ignoring NEUTRAL. **Rice is the weakest and it is the product with the most examples**, which is a real open problem rather than a rounding artefact.

9.4 Rolling origin backtest

Retrains on everything before each quarter and tests on that quarter, so it is not one lucky split. This is the more trustworthy evidence, because the single test split has only 40 rows.

Quarter	Model	Majority	Test rows
2024Q3	0.259	0.364	14
2024Q4	0.331	0.296	25
2025Q1	0.351	0.173	40
2025Q2	0.235	0.253	31
2025Q3	0.423	0.202	39
2025Q4	0.423	0.175	14
2026Q1	0.167	0.200	21
2026Q2	0.287	0.237	20
2026Q3	0.247	0.429	20

Beats the majority baseline in 5 of 9 quarters.

10. Decision support: urgency and action

A direction and a probability is not a decision. Each prediction is scored for urgency out of 100 from four components, then mapped to a priority and a concrete next step.

Component	Weight	What it measures
Model confidence	40	scaled by the predicted probability
Direction cost	20	a rise costs the buyer money and scores full; a fall merely saves it and scores half
Price corroboration	25	full when the official 7 day move already agrees and exceeds 0.5 percent, partial when it agrees weakly, and a recorded caution when it disagrees
Recency	15	full within 24 hours, partial to 72 hours, none beyond

Two caps stop a coin flip reaching the top of the list. ACT NOW requires the confidence band that actually tested at 70 percent correct; anything less reaches WATCH at most. NEUTRAL never exceeds NO ACTION, because carry on as normal is not an alert however confident the model is.

Score	Priority	Example action
70+	ACT NOW	Buy your next chicken stock now, before the increase reaches the shelf
45 to 69	WATCH	Hold off on buying beyond what you need now, prices look likely to ease
25 to 44	MONITOR	Check prices at your usual supplier this week
below 25	NO ACTION	Buy as normal

11. Explainability: how news reaches a shelf price

26 documented mechanisms turn a detected event word into a plain causal chain: what the event is, why it moves prices, which way, and how long it takes. Covering shortages, floods, drought, bird flu, feed costs, import and export bans, tariffs, tax, the ringgit, festive demand, subsidies and price controls.

```
1. The article talks about chicken (it mentions chicken, ayam).
2. It mentions "feed cost": the cost of animal feed is rising. Feed is the largest
   single cost in raising poultry, so farmers pass it on, pushing prices upward.
3. Official PriceCatcher data shows chicken prices already rising (+1.4% over 7 days).
4. Any change usually reaches the shelf in 1 to 2 weeks.
```

Two honesty constraints are built in. Every output states that this is the established domain relationship and *not* a readout of what the model weighted internally; claiming otherwise would be inventing an explanation. And disagreement is reported rather than smoothed: when the mechanism points the opposite way to the model, the note says to trust the alert less.

The explainer that writes sentences receives only computed facts under a strict schema, and by default the chat host writes the prose while the server writes none. The server therefore cannot invent a number, because it never writes a sentence containing one.

12. The MCP interface

Tool	Inputs	Returns
list_domains	none	the domain, its four products, aliases, horizon
fetch_news	hours, sources, limit	recent headlines from the configured feeds
filter_relevant	domain, hours, limit	articles that survive all three filter stages, with scores
predict_impact	url or text or title, products	impact label, probabilities, confidence band, triggers, causal reasoning
price_context	product, days	latest index, 7 and 30 day change, series tail, data_as_of
scan_alerts	domain, hours, min_confidence, lang, limit	the main view: ranked alerts with priority, action, reasoning and price context
backtest_report	product	test metrics, three baselines, interval, per product, rolling origin, label legend
forward_validation	product	live predictions that have matured, scored against real prices
demo_cases	right, wrong	held out cases where the real outcome is known, including incorrect ones

Every tool returns JSON. On failure it returns `{"error": ..., "hint": ...}` and never raises to the client. Logging goes to stderr only, because stdout is the protocol channel and one stray print corrupts it; a test enforces this.

Out of domain requests are declined, not guessed. An unknown product returns an error naming the four valid ones. An article about Bitcoin or petrol returns "No tracked food staple is discussed in this article". Nonsense input does the same.

12.1 Deployment

Transport	Use	Notes
stdio	Claude Desktop, local clients	The default and the most reliable. Configured in claude_desktop_config.json
streamable HTTP	remote clients, tunnels	--http with --host, --port, --path and --allow-host. Runs stateless so each request is independent through a proxy

--allow-host exists for a real trap. The SDK rejects any request whose Host header it does not recognise, which is correct DNS rebinding protection but blocks every tunnelled request, since a tunnel hostname is not localhost.

Cloudflare tunnels need outbound port 7844, which the development network blocks in both TCP and UDP. localtunnel and VS Code port forwarding work over 443 and are documented as the fallbacks.

13. Testing

145 tests. The suite is written to pin down behaviour that was wrong at some point, not to chase coverage.

Area	What is enforced
Alias filter	word boundaries, Malay aliases, exclude phrases masking only their own span
Price index	outlier trimming keeps a genuine step change, normalisation, weekly survey carry forward
Labels	threshold boundaries, theta from train only, the purge gap actually holding
Relevance	the price cue rule, degradation when the embedding model is absent, both stages reading one config

Area	What is enforced
Reasoning	mechanisms explained not listed, disagreement flagged, no mechanism admitted rather than invented
Action	ACT NOW needs high confidence, NEUTRAL never urgent, stale news ranked lower
Tools	every tool returns JSON, the error shape, no stdout writes, one vocabulary
End to end	the full offline pipeline on synthetic data, so the code is known good before real data

14. Limitations, stated plainly

These are the things a careful reader should know. They are in the README and the tool output too, not only here.

- **Only 305 training examples** from 153,981 articles. The Star publishes few staple price stories. This is the binding constraint on everything else.
- **The margin over momentum is 0.018**, inside the noise. For raw accuracy you could use momentum alone. What momentum cannot do is say why, surface the article, or flag a subsidy announcement the day it lands.
- **The test set has been evaluated several times** across configurations. Selection was done on validation, but repeated looks make the headline figure optimistic by an unknown margin. The rolling origin numbers are more trustworthy.
- **Rice has the most examples and the worst directional accuracy.** Unexplained.
- **The model over-predicts NEUTRAL** and can be confidently wrong. One demo case is a 95 percent confident NEUTRAL that was wrong.
- **Typos defeat the alias stage.** Matching is exact, so "chiken" does not match. Fuzzy matching is a known, unimplemented fix.
- **Training news is one source.** Trained on The Star, run live on Malay Mail, Bernama and FMT, so live performance is an out of source test.
- **Electronics was removed** rather than validated badly.
- **The item mapping is provisional**, ranked by history and coverage rather than reviewed by a domain expert.
- **Forward validation has no matured rows yet.** The mechanism works; the first scoreable prediction is 28 days after the daily job starts.

15. What would actually improve it

Priority	Change	Expected effect
1	A broader news corpus beyond one outlet	Directly attacks the binding constraint. Everything else is downstream of example count
2	A third labelled audit, single annotator, on this pipeline's own candidate pool	The current precision estimate rests on two partly incompatible audits
3	Fuzzy alias matching	Cheap. rapidfuzz is already a dependency
4	Investigate rice specifically	Most examples, worst result. Either the labels or the filter are wrong for it
5	Per product models, once data allows	Merging was forced by data volume, not chosen on principle

16. Appendix: configuration

Parameter	Value	Meaning
horizon_days	28	days ahead the label looks
theta_k	0.5	multiplier on the training standard deviation
MIN_THETA	0.005	threshold floor for price controlled products
min_premises	5	a day with fewer reporting premises is dropped
embed_threshold	0.015	contrastive relevance cut off
requires_price_cue	True	a cue word from 28 is required in the headline

Parameter	Value	Meaning
min_confidence	0.55	below this the model abstains
alert_confidence	0.7	the band that tested at 70 percent
body_chars	1500	body characters the alias stage reads
rate limit	1 request per second per host	never lowered; robots.txt obeyed

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